NGO BA TAN TAI

Software Engineer — Backend

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Portfolio: https://taingo2112003.github.io/NBTT_Resume/ | GitHub: github.com/TaiNgo2112003

TECHNICAL SKILLS

- **Backend Architecture & Core:** C# (.NET Core, ASP.NET Core, .NET 8/9/10, ABP Framework, Entity Framework Core), TypeScript, Node.js (NestJS 11, Express), Python (FastAPI). Core specialization in enterprise C# / .NET backend systems, Clean Architecture, CQRS, Modular Monolith & Microservices.
- **Distributed Systems & Messaging:** gRPC, Apache Kafka, RabbitMQ, Redis (ioredis, BullMQ), Outbox/Inbox Pattern, WebSockets (Socket.IO).
- **Databases & Observability:** SQL Server, PostgreSQL 14+ (TypeORM 0.3, EF Core), MongoDB, Redis | Prometheus, Grafana, Loki, Winston Structured Logging.
- **DevOps, Cloud & Infrastructure:** Docker, Docker Compose, Linux VPS, Nginx, PgBouncer (Transaction pooling), Cloudflare (Load Balancing, R2), AWS S3, Self-Hosted GitHub Actions CI/CD automation.
- **Security & Authentication:** JWT / Passport, OpenIddict, SmartOTP V2 (EC P-256, Anti-Hooking Frida/Xposed), PIN Step-up Tokens, VNPT / FPT eKYC, HMAC-SHA512, CSRF, Helmet, Gitleaks.
- **Testing & Tooling:** xUnit, Jest, Supertest, Testcontainers (isolated Postgres/Redis), k6 smoke/load-test, Git, Swagger/OpenAPI, Husky, ESLint.
- **AI & Machine Learning (Secondary):** Local LLM deployment & acceleration (Context Window management, KV Cache, RAG), ONNX Runtime (InsightFace, ViT, PhoBERT-v2 HSD), Scikit-Learn, Dynamic Quantization (INT8), FFmpeg.

PROFESSIONAL EXPERIENCE

Vector T — Backend Engineer — 05/2026 – Present

TrustWow Platform — Social-Trust & Marketplace Ecosystem

Phase 1: High-Performance Modular Monolith (NestJS 11, PostgreSQL, Redis)

- **Financial Ledger Engine & Payment Gateway:** Architected an append-only double-entry accounting ledger system, accounting toolkit, and automated payout settlement pipelines; integrated VNPay payment gateway with HMAC-SHA512 signature verification, IPN webhook handlers, QueryDR reconciliation, transaction idempotency keys, and automated financial audit logging.
- **Security Engineering, SmartOTP V2 & Cryptography:** Engineered SmartOTP V2 leveraging device-bound Elliptic Curve cryptography (EC P-256 challenge-response), anti-hooking detection (Frida, Xposed runtime guards), and cross-platform push approval protocols for web sessions; implemented robust JWT/Passport authentication (Bearer tokens & HttpOnly cookies), TypeORM encrypted PII columns, and VNPT eKYC biometric verification.
- **Realtime Infrastructure, WebRTC & Media Processing:** Built scalable realtime communication infrastructure using WebSocket (Socket.IO) backed by Redis Pub/Sub for state synchronization and broadcast messaging; deployed and tuned Coturn STUN/TURN servers for WebRTC peer-to-peer audio/video calling; integrated native FFmpeg worker pipelines for asynchronous video thumbnail processing streaming directly to Cloudflare R2 object storage.
- **Modular Architecture, Testing & High-Throughput Engineering:** Maintained strict TypeScript modular monolith architecture on NestJS 11 and PostgreSQL; implemented complex database migrations and custom repositories in TypeORM 0.3; enforced automated testing and code quality using Jest, Supertest, Testcontainers (isolated Postgres/Redis containers), fast-check property-based testing, and Husky/ESLint git hooks.
- **AI, Machine Learning & Content Moderation (Sidecars):** Engineered standalone Python FastAPI microservices and background pipelines: built an AI Content Moderation Sidecar integrating PhoBERT-v2 (visolex/phobert-v2-hsd) for Vietnamese text moderation (sliding-window 238-token chunking) and ViT (OwenElliott/image-safety-classifier-xs) for NSFW/safety detection, optimized via INT8 Dynamic Quantization and BullMQ async queues for sub-second moderation on CPU instances; developed a personalized News Feed Recommendation Sidecar (Logistic Regression across 5 engagement dimensions) with model weights exported to Cloudflare R2 for zero-latency native TypeScript inference (sigmoid(dot(features, coefficients))); integrated VNPT eKYC and local InsightFace Buffalo_s (det_500m.onnx, w600k_mbf.onnx) with TTA, landmark ROI crop, and FaceVerificationFreshGuard for biometric transaction step-up; authored custom GPS Anti-Spoofing algorithm (5-sample cadence, velocity, micro-jitter analysis) blocking Mock Location / FakeGPS spoofing.
- **DevOps, Cloud Infrastructure & Observability:** Architected Self-Hosted GitHub Actions Runners (Linux VPS & internal Mac mini) for PR CI pipelines and automated multi-environment deployment (Develop/Staging), cutting 100% of GitHub Actions billable minutes; migrated production Docker environment to Debian Bookworm-slim to support native C++ ONNX/OpenCV runtimes; orchestrated backend, sidecars, and Nginx reverse proxy via Docker Compose with local layer caching, automated SSL, health checks, and zero-downtime deploy scripts; configured PgBouncer (Transaction pooling mode) in front of PostgreSQL to eliminate max_connections bottlenecks and slash database RAM usage; built full-stack observability using Prometheus (custom NestJS metrics interceptors for latency, RPS, memory), Grafana dashboards, Loki log aggregation, and Winston structured audit logging.

Phase 2: Microservices Decomposition & High Availability (Architecture & Transition)

- **Microservices Decomposition & gRPC:** Strategically decoupled core domain boundaries from the Modular Monolith into dedicated standalone microservices (Auth & Identity, Payment & Ledger, Chat & Messaging, and Media & Storage); standardized high-throughput inter-service Remote Procedure Calls via gRPC (Protobuf), while utilizing BullMQ with Redis for reliable, asynchronous cross-service job orchestration and event-driven workload distribution.
- **High Availability & Cloudflare Load Balancing:** Architected a multi-tier resilient infrastructure via Cloudflare Load Balancing—routing primary production traffic to a dedicated high-capacity primary server, while orchestrating secondary VPS nodes for auxiliary load distribution and active-standby automated failover backup during unexpected downtime or peak traffic bursts.

Savills — Backend Engineer **07/2025 – 05/2026**

Real Estate Management Platform — Microservices Architecture, Multi-tenant, Multi-national (C#, .NET Core, ABP Framework, SQL Server)

- Collaborated with cross-regional Southeast Asia engineering teams to design and develop backend microservices for Savills' enterprise real estate and resident management platform using C# and ABP Framework.
- Built a scalable, secure multi-tenant architecture supporting property and resident operations across multiple regions; seamlessly integrated with external payment gateways and banking services (ZaloPay, MoMo, HDBank, Techcombank).
- Engineered an anonymous resident feedback system with CAPTCHA verification to eliminate spam; integrated with FacilityBot API to automate property management workflows and maintenance service request lifecycles.
- **Regional Multi-Tenant Payments:** Developed tenant-aware payment workflows for the platform operating in Singapore and Thailand, integrating additional banking providers to support regional payment requirements.
- **ML Model Training for Post Categorization:** Directly trained a machine-learning model to predict categories for platform posts, supporting automated content classification; integrated AI assistant capabilities via Microsoft Copilot to optimize internal administrative workflows.

Titan Technology — Fullstack Developer / AI Engineer **03/2025 – 07/2025**

Logic-Based Learning Systems & Internal AI (Python, Flask, SWI-Prolog, Node.js, SQL Server, PostgreSQL)

- **Internal Local AI Integration:** Researched, deployed, and integrated an on-premise Local AI system for company-wide internal usage; optimized large prompt execution and response latency by implementing custom Context Window management and KV Cache acceleration; architected an enterprise RAG (Retrieval-Augmented Generation) pipeline connecting local vector databases to internal codebases and technical documentation for rapid, hallucination-free knowledge retrieval.
- Contributed to backend API architecture and frontend interfaces for inductive logic programming (ILP) and machine learning platforms; developed web applications for structured data ingestion and custom PyGol-like evaluation pipelines.
- **Engagement Context:** Contractor role focused on maintaining and updating an existing company project and its backend/frontend codebase.

PATSOFT — Software Developer **07/2023 – 06/2024**

Enterprise ERP & Logistics Systems (C#, .NET Framework, .HQ Internal Framework, ReactJS, EXT JS, SQL Server)

- **Enterprise ERP Architecture:** Engineered enterprise warehouse, inventory, human resources (HR), and supply chain logistics management modules using C# and .NET on the company's proprietary .HQ framework.
- **MISA E-Invoicing Integration (GTGT / VAT):** Architected direct system integration with the MISA API for electronic Value-Added Tax (VAT / GTGT) invoice generation, XML data schema validation, digital signing, and regulatory tax synchronization, ensuring strict compliance with Vietnamese e-invoicing standards.
- **Public Service Digitalization:** Integrated hardware barcode scanning systems; partnered with provincial public administration units across the Mekong Delta on the 'ATM Document Withdrawal' initiative to digitize and automate citizen record retrieval.
- **Payment Integration & Database Optimization:** Developed payment processing functionality and product catalog management applications serving retail branches across Ho Chi Minh City; optimized SQL Server queries, indexing, and stored procedures for high-volume transactions.

EDUCATION & CERTIFICATIONS

HCM University of Natural Resources and Environment **2021 – 2025**

Bachelor of Software Engineering — GPA: 3.12 (Full Scholarship for 3 consecutive years) | TOEIC: 630

FEATURED PROJECTS

Learning English System — Personal Project **GitHub: Learning-English-System**

Tech: C# / .NET 10, EF Core, SQL Server, OpenIddict, Kafka, gRPC, Redis, Gemini, React, TypeScript, Docker

Repository URL: https://github.com/orgs/Learning-English-System/repositories

- **Service Architecture & Integration:** Structured five backend services (Auth, Profile, Learning, AI, and Test) across seven repositories with separate frontend and infrastructure; implemented SQL-backed outbox/inbox adapters, Kafka registration events, and gRPC clients for learning/test/AI integration.
- **Adaptive Learning Algorithm:** Implemented SM-2 spaced repetition with recall grades, ease factors, lapse tracking, review intervals, and mastery thresholds, with unit tests for scheduling behavior.
- **Identity & AI Workflows:** Implemented OpenIddict token flows, refresh-token family/reuse checks, and session revocation; integrated Gemini for vocabulary analysis and quiz distractor generation with Redis response caching.
- **Frontend, Infrastructure & Testing:** Built a React learning interface, Nginx gateway, and Docker Compose tooling; added GitHub Actions build/test workflows, backend unit/integration suites, and k6 smoke/load-test scripts.

MY RESUME / PORTFOLIO

SCAN QR CODE OR CLICK ON THE FOLLOWING LINK TO LEARN MORE ABOUT ME:

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